principles, so do we in work of every kind. For the living thing is generated before the animal, and the animal before the man, as is shown in De Gener. Anim. ii, 3. It was fitting, then, as an evidence of the Divine wisdom, that among the works of distinction the production of light should take first place, since light is a form of the primary body, and because it is more common quality.

Basil [*Hom. ii in Hexaem.], indeed, adds a third reason: that all other things are made manifest by light. And there is yet a fourth, already touched upon in the objections; that day cannot be unless light exists, which was made therefore on the first day.

Reply to Objection 1: According to the opinion of those who hold that the formlessness of matter preceded its form in duration, matter must be held to have been created at the beginning with substantial forms, afterwards receiving those that are accidental, among which light holds the first place.

Reply to Objection 2: In the opinion of some the light here spoken of was a kind of luminous nebula, and that on the making of the sun this returned to the matter of which it had been formed. But this cannot well be maintained, as in the beginning of Genesis Holy Scripture records the institution of that order of nature which henceforth is to endure. We cannot, then, say that what was made at that time afterwards ceased to exist.

Others, therefore, held that this luminous nebula continues in existence, but so closely attached to the sun as to be indistinguishable. But this is as much as to say that it is superfluous, whereas none of God's works have been made in vain. On this account it is held by some that the sun's body was made out of this nebula. This, too, is impossible to those at least who believe that the sun is different in its nature from the four elements, and naturally incorruptible. For in that case its matter cannot take on another form.

I answer, then, with Dionysius (Div. Nom. iv), that the light was the sun's light, formless as yet, being already the solar substance, and possessing illuminative power in a general way, to which was afterwards added the special and determinative power required to produce determinate effects. Thus, then, in the production of this light a triple distinction was made between light and darkness. First, as to the cause, forasmuch as in the substance of the sun we have the cause of light, and in the opaque nature of the earth the cause of darkness. Secondly, as to place, for in one hemisphere there was light, in the other darkness. Thirdly, as to time; because there was light for one and darkness for another in the same hemisphere; and this is signified by the words, "He called the light day, and the darkness night."

Reply to Objection 3: Basil says (Hom. ii in Hexaem.) that day and night were then caused by expansion and contraction of light, rather than by movement. But Augustine objects to this (Gen. ad lit. i), that there was no reason for this vicissitude of expansion and contraction since there were neither men nor animals on the earth at that time, for whose service this was required. Nor does the nature of a luminous body seem to admit of the withdrawal of light, so long as the body is actually present; though this might be effected by a miracle. As to this, however, Augustine remarks (Gen. ad lit. i) that in the first founding of the order of nature we must not look for miracles, but for what is in accordance with nature. We hold, then, that the movement of the heavens is twofold. Of these movements, one is common to the entire heaven, and is the cause of day and night. This, as it seems, had its beginning on the first day. The other varies in proportion as it affects various bodies, and by its variations is the cause of the succession of days, months, and years. Thus it is, that in the account of the first day the distinction between day and night alone is mentioned; this distinction being brought about by the common movement of the heavens. The further distinction

into successive days, seasons, and years recorded as begun on the fourth day, in the words, "let them be for seasons, and for days, and years" is due to proper movements.

Reply to Objection 4: As Augustine teaches (Confess. xii; Gen. ad lit. 1,15), formlessness did not precede forms in duration; and so we must understand the production of light to signify the formation of spiritual creatures, not, indeed, with the perfection of glory, in which they were not created, but with the perfection of grace, which they possessed from their creation as said above (Q[62], A[3]). Thus the division of light from darkness will denote the distinction of the spiritual creature from other created things as yet without form. But if all created things received their form at the same time, the darkness must be held to mean the spiritual darkness of the wicked, not as existing from the beginning but such as God foresaw would exist.

ON THE WORK OF THE SECOND DAY (FOUR ARTICLES)

We must next consider the work of the second day. Under this head there are four points of inquiry:

(1) Whether the firmament was made on the second day?

(2) Whether there are waters above the firmament?

(3) Whether the firmament divides waters from waters?

(4) Whether there is more than one heaven?

Whether the firmament was made on the second day?

Objection 1: It would seem that the firmament was not made on the second day. For it is said (Gn. 1:8): "God called the firmament heaven." But the heaven existed before days, as is clear from the words, "In the beginning God created heaven and earth." Therefore the firmament was not made on the second day.

Objection 2: Further, the work of the six days is ordered conformably to the order of Divine wisdom. Now it would ill become the Divine wisdom to make afterwards that which is naturally first. But though the firmament naturally precedes the earth and the waters, these are mentioned before the formation of light, which was on the first day. Therefore the firmament was not made on the second day.

Objection 3: Further, all that was made in the six days was formed out of matter created before days began. But the firmament cannot have been formed out of pre-existing matter, for if so it would be liable to generation and corruption. Therefore the firmament was not made on the second day.

On the contrary, It is written (Gn. 1:6): "God said: let there be a firmament," and further on (verse 8); "And the evening and morning were the second day."

I answer that, In discussing questions of this kind two rules are to observed, as Augustine teaches (Gen. ad lit. i, 18). The first is, to hold the truth of Scripture without wavering. The second is that since Holy Scripture can be explained in a multiplicity of senses, one should adhere to a particular explanation, only in such measure as to be ready to abandon it, if it be proved with certainty to be false; lest Holy Scripture be exposed to the ridicule of unbelievers, and obstacles be placed to their believing.

We say, therefore, that the words which speak of the firmament as made on the second day can be understood in two senses. They may be understood, first, of the starry firmament, on which point it is necessary to set forth the different opinions of philosophers. Some of these believed it to be composed of the elements; and this was the opinion of Empedocles, who, however, held further that the body of the firmament was not susceptible of dissolution, because its parts are, so to say, not in disunion, but in harmony. Others held the firmament to be of the nature of the four elements, not, indeed, compounded of them, but being as it were a simple element. Such was the opinion of Plato, who held that element to be fire. Others, again, have held that the heaven is not of the nature of the four elements, but is itself a fifth body, existing over and above these. This is the opinion of Aristotle (De Coel. i, text. 6,32).

According to the first opinion, it may, strictly speaking, be granted that the firmament was made, even as to substance, on the second day. For it is part of the work of creation to produce the substance of the elements, while it belongs to the work of distinction and adornment to give forms to the elements that pre-exist.

But the belief that the firmament was made, as to its substance, on the second day is incompatible with the opinion of Plato, according to whom the making of the firmament implies the production of the element of fire. This production, however, belongs to the work of creation, at least, according to those who hold that formlessness of matter preceded in time its formation, since the first form received by matter is the elemental.

Still less compatible with the belief that the substance of the firmament was produced on the second day is the opinion of Aristotle, seeing that the mention of days denotes succession of time, whereas the firmament, being naturally incorruptible, is of a matter not susceptible of change of form; wherefore it could not be made out of matter existing antecedently in time.

Hence to produce the substance of the firmament belongs to the work of creation. But its formation, in some degree, belongs to the second day, according to both opinions: for as Dionysius says (Div. Nom. iv), the light of the sun was without form during the first three days, and afterwards, on the fourth day, received its form.

If, however, we take these days to denote merely sequence in the natural order, as Augustine holds (Gen. ad lit. iv, 22,24), and not succession in time, there is then nothing to prevent our saying, whilst holding any one of the opinions given above, that the substantial formation of the firmament belongs to the second day.

Another possible explanation is to understand by the firmament that was made on the second day, not that in which the stars are set, but the part of the atmosphere where the clouds are collected, and which has received the name firmament from the firmness and density of the air. "For a body is called firm," that is dense and solid, "thereby differing from a mathematical body" as is remarked by Basil (Hom. iii in Hexaem.). If, then, this explanation is adopted none of these opinions will be found repugnant to reason. Augustine, in fact (Gen. ad lit. ii, 4), recommends it thus: "I consider this view of the question worthy of all commendation, as neither contrary to faith nor difficult to be proved and believed."

Reply to Objection 1: According to Chrysostom (Hom. iii in Genes.), Moses prefaces his record by speaking of the works of God collectively, in the words, "In the beginning God created heaven and earth," and then proceeds to explain them part by part; in somewhat the same way as one might say: "This house was constructed by that builder," and then add: "First, he laid the foundations, then built the walls, and thirdly, put on the roof." In accepting this explanation we are,

therefore, not bound to hold that a different heaven is spoken of in the words: "In the beginning God created heaven and earth," and when we read that the firmament was made on the second day.

We may also say that the heaven recorded as created in the beginning is not the same as that made on the second day; and there are several senses in which this may be understood. Augustine says (Gen. ad lit. i, 9) that the heaven recorded as made on the first day is the formless spiritual nature, and that the heaven of the second day is the corporeal heaven. According to Bede (Hexaem. i) and Strabus, the heaven made on the first day is the empyrean, and the firmament made on the second day, the starry heaven. According to Damascene (De Fide Orth. ii) that of the first day was spherical in form and without stars, the same, in fact, that the philosophers speak of, calling it the ninth sphere, and the primary movable body that moves with diurnal movement: while by the firmament made on the second day he understands the starry heaven. According to another theory, touched upon by Augustine [*Gen. ad lit. ii, 1] the heaven made on the first day was the starry heaven, and the firmament made on the second day was that region of the air where the clouds are collected, which is also called heaven, but equivocally. And to show that the word is here used in an equivocal sense, it is expressly said that "God called the firmament heaven"; just as in a preceding verse it said that "God called the light day" (since the word "day" is also used to denote a space of twenty-four hours). Other instances of a similar use occur, as pointed out by Rabbi Moses.

The second and third objections are sufficiently answered by what has been already said.

Whether there are waters above the firmament?

Objection 1: It would seem that there are not waters above the firmament. For water is heavy by nature, and heavy things tend naturally downwards, not upwards. Therefore there are not waters above the firmament.

Objection 2: Further, water is fluid by nature, and fluids cannot rest on a sphere, as experience shows. Therefore, since the firmament is a sphere, there cannot be water above it.

Objection 3: Further, water is an element, and appointed to the generation of composite bodies, according to the relation in which imperfect things stand towards perfect. But bodies of composite nature have their place upon the earth, and not above the firmament, so that water would be useless there. But none of God's works are useless. Therefore there are not waters above the firmament.

On the contrary, It is written (Gn. 1:7): "(God) divided the waters that were under the firmament, from those that were above the firmament."

I answer with Augustine (Gen. ad lit. ii, 5) that, "These words of Scripture have more authority than the most exalted human intellect. Hence, whatever these waters are, and whatever their mode of existence, we cannot for a moment doubt that they are there." As to the nature of these waters, all are not agreed. Origen says (Hom. i in Gen.) that the waters that are above the firmament are "spiritual substances." Wherefore it is written (Ps. 148:4): "Let the waters that are above the heavens praise the name of the Lord," and (Dan. 3:60): "Ye waters that are above the heavens, bless the Lord."To this Basil answers (Hom. iii in Hexaem.) that these words do not mean that these waters are rational creatures, but that "the thoughtful contemplation of them by those who understand fulfils the glory of the Creator." Hence in the same context, fire, hail, and other like creatures, are invoked in the same way, though no one would attribute reason to these.

We must hold, then, these waters to be material, but their exact nature will be differently defined according as opinions on the firmament differ. For if by the firmament we understand the starry heaven, and as being of the nature of the four elements, for the same reason it may be believed that the waters above the heaven are of the same nature as the elemental waters. But if by the firmament we understand the starry heaven, not, however, as being of the nature of the four elements then the waters above the firmament will not be of the same nature as the elemental waters, but just as, according to Strabus, one heaven is called empyrean, that is, fiery, solely on account of its splendor: so this other heaven will be called aqueous solely on account of its transparence; and this heaven is above the starry heaven. Again, if the firmament is held to be of other nature than the elements, it may still be said to divide the waters, if we understand by water not the element but formless matter. Augustine, in fact, says (Super Gen. cont. Manich. i, 5,7) that whatever divides bodies from bodies can be said to divide waters from waters.

If, however, we understand by the firmament that part of the air in which the clouds are collected, then the waters above the firmament must rather be the vapors resolved from the waters which are raised above a part of the atmosphere, and from which the rain falls. But to say, as some writers alluded to by Augustine (Gen. ad lit. ii, 4), that waters resolved into vapor may be lifted above the starry heaven, is a mere absurdity. The solid nature of the firmament, the intervening region of fire, wherein all vapor must be consumed, the tendency in light and rarefied bodies to drift to one spot beneath the vault of the moon, as well as the fact that vapors are perceived not to rise even to the tops of the higher mountains, all to go to show the impossibility of this. Nor is it less absurd to say, in support of this opinion, that bodies may be rarefied infinitely, since natural bodies cannot be infinitely rarefied or divided, but up to a certain point only.

Reply to Objection 1: Some have attempted to solve this difficulty by supposing that in spite of the natural gravity of water, it is kept in its place above the firmament by the Divine power. Augustine (Gen. ad lit. ii, 1), however will not admit this solution, but says "It is our business here to inquire how God has constituted the natures of His creatures, not how far it may have pleased Him to work on them by way of miracle." We leave this view, then, and answer that according to the last two opinions on the firmament and the waters the solution appears from what has been said. According to the first opinion, an order of the elements must be supposed different from that given by Aristotle, that is to say, that the waters surrounding the earth are of a dense consistency, and those around the firmament of a rarer consistency, in proportion to the respective density of the earth and of the heaven.

Or by the water, as stated, we may understand the matter of bodies to be signified.

Reply to Objection 2: The solution is clear from what has been said, according to the last two opinions. But according to the first opinion, Basil gives two replies (Hom. iii in Hexaem.). He answers first, that a body seen as concave beneath need not necessarily be rounded, or convex, above. Secondly, that the waters above the firmament are not fluid, but exist outside it in a solid state, as a mass of ice, and that this is the crystalline heaven of some writers.

Reply to Objection 3: According to the third opinion given, the waters above the firmament have been raised in the form of vapors, and serve to give rain to the earth. But according to the second opinion, they are above the heaven that is wholly transparent and starless. This, according to some, is the primary mobile, the cause of the daily revolution of the entire heaven, whereby the continuance of generation is secured. In the same way the starry heaven, by the zodiacal movement, is the cause whereby different bodies are generated or corrupted, through the rising and setting of

the stars, and their various influences. But according to the first opinion these waters are set there to temper the heat of the celestial bodies, as Basil supposes (Hom. iii in Hexaem.). And Augustine says (Gen. ad lit. ii, 5) that some have considered this to be proved by the extreme cold of Saturn owing to its nearness to the waters that are above the firmament.

Whether the firmament divides waters from waters?

Objection 1: It would seem that the firmament does not divide waters from waters. For bodies that are of one and the same species have naturally one and the same place. But the Philosopher says (Topic. i, 6): "All water is the same species." Water therefore cannot be distinct from water by place.

Objection 2: Further, should it be said that the waters above the firmament differ in species from those under the firmament, it may be argued, on the contrary, that things distinct in species need nothing else to distinguish them. If then, these waters differ in species, it is not the firmament that distinguishes them.

Objection 3: Further, it would appear that what distinguishes waters from waters must be something which is in contact with them on either side, as a wall standing in the midst of a river. But it is evident that the waters below do not reach up to the firmament. Therefore the firmament does not divide the waters from the waters.

On the contrary, It is written (Gn. 1:6): "Let there be a firmament made amidst the waters; and let it divide the waters from the waters."

I answer that, The text of Genesis, considered superficially, might lead to the adoption of a theory similar to that held by certain philosophers of antiquity, who taught that water was a body infinite in dimension, and the primary element of all bodies. Thus in the words, "Darkness was upon the face of the deep," the word "deep" might be taken to mean the infinite mass of water, understood as the principle of all other bodies. These philosophers also taught that not all corporeal things are confined beneath the heaven perceived by our senses, but that a body of water, infinite in extent, exists above that heaven. On this view the firmament of heaven might be said to divide the waters without from those within---that is to say, from all bodies under the heaven, since they took water to be the principle of them all.

As, however, this theory can be shown to be false by solid reasons, it cannot be held to be the sense of Holy Scripture. It should rather be considered that Moses was speaking to ignorant people, and that out of condescension to their weakness he put before them only such things as are apparent to sense. Now even the most uneducated can perceive by their senses that earth and water are corporeal, whereas it is not evident to all that air also is corporeal, for there have even been philosophers who said that air is nothing, and called a space filled with air a vacuum.

Moses, then, while he expressly mentions water and earth, makes no express mention of air by name, to avoid setting before ignorant persons something beyond their knowledge. In order, however, to express the truth to those capable of understanding it, he implies in the words: "Darkness was upon the face of the deep," the existence of air as attendant, so to say, upon the water. For it may be understood from these words that over the face of the water a transparent body was extended, the subject of light and darkness, which, in fact, is the air.

Whether, then, we understand by the firmament the starry heaven, or the cloudy region of the air, it is true to say that it divides the waters from the waters, according as we take water to denote formless matter, or any kind of transparent body, as fittingly designated under the name of waters. For the starry heaven divides the lower transparent bodies from the higher, and the cloudy region divides that higher part of the air, where the rain and similar things are generated, from the lower part, which is connected with the water and included under that name.

Reply to Objection 1: If by the firmament is understood the starry heaven, the waters above are not of the same species as those beneath. But if by the firmament is understood the cloudy region of the air, both these waters are of the same species, and two places are assigned to them, though not for the same purpose, the higher being the place of their begetting, the lower, the place of their repose.

Reply to Objection 2: If the waters are held to differ in species, the firmament cannot be said to divide the waters, as the cause of their destruction, but only as the boundary of each.

Reply to Objection 3: On account of the air and other similar bodies being invisible, Moses includes all such bodies under the name of water, and thus it is evident that waters are found on each side of the firmament, whatever be the sense in which the word is used.

Whether there is only one heaven?

Objection 1: It would seem that there is only one heaven. For the heaven is contrasted with the earth, in the words, "In the beginning God created heaven and earth."But there is only one earth. Therefore there is only one heaven.

Objection 2: Further, that which consists of the entire sum of its own matter, must be one; and such is the heaven, as the Philosopher proves (De Coel. i, text. 95). Therefore there is but one heaven.

Objection 3: Further, whatever is predicated of many things univocally is predicated of them according to some common notion. But if there are more heavens than one, they are so called univocally, for if equivocally only, they could not properly be called many. If, then, they are many, there must be some common notion by reason of which each is called heaven, but this common notion cannot be assigned. Therefore there cannot be more than one heaven.

On the contrary, It is said (Ps. 148:4): "Praise Him, ye heavens of heavens."

I answer that, On this point there seems to be a diversity of opinion between Basil and Chrysostom. The latter says that there is only one heaven (Hom. iv in Gen.), and that the words 'heavens of heavens' are merely the translation of the Hebrew idiom according to which the word is always used in the plural, just as in Latin there are many nouns that are wanting in the singular. On the other hand, Basil (Hom. iii in Hexaem.), whom Damascene follows (De Fide Orth. ii), says that there are many heavens. The difference, however, is more nominal than real. For Chrysostom means by the one heaven the whole body that is above the earth and the water, for which reason the birds that fly in the air are called birds of heaven [*Ps. 8:9]. But since in this body there are many distinct parts, Basil said that there are more heavens than one.

In order, then, to understand the distinction of heavens, it must be borne in mind that Scripture speaks of heaven in a threefold sense. Sometimes it uses the word in its proper and natural meaning, when it denotes that body on high which is luminous actually or potentially, and incorruptible by

nature. In this body there are three heavens; the first is the empyrean, which is wholly luminous; the second is the aqueous or crystalline, wholly transparent; and the third is called the starry heaven, in part transparent, and in part actually luminous, and divided into eight spheres. One of these is the sphere of the fixed stars; the other seven, which may be called the seven heavens, are the spheres of the planets.

In the second place, the name heaven is applied to a body that participates in any property of the heavenly body, as sublimity and luminosity, actual or potential. Thus Damascene (De Fide Orth. ii) holds as one heaven all the space between the waters and the moon's orb, calling it the aerial. According to him, then, there are three heavens, the aerial, the starry, and one higher than both these, of which the Apostle is understood to speak when he says of himself that he was "rapt to the third heaven."

But since this space contains two elements, namely, fire and air, and in each of these there is what is called a higher and a lower region Rabanus subdivides this space into four distinct heavens. The higher region of fire he calls the fiery heaven; the lower, the Olympian heaven from a lofty mountain of that name: the higher region of air he calls, from its brightness, the ethereal heaven; the lower, the aerial. When, therefore, these four heavens are added to the three enumerated above, there are seven corporeal heavens in all, in the opinion of Rabanus.

Thirdly, there are metaphorical uses of the word heaven, as when this name is applied to the Blessed Trinity, Who is the Light and the Most High Spirit. It is explained by some, as thus applied, in the words, "I will ascend into heaven"; whereby the evil spirit is represented as seeking to make himself equal with God. Sometimes also spiritual blessings, the recompense of the Saints, from being the highest of all good gifts, are signified by the word heaven, and, in fact, are so signified, according to Augustine (De Serm. Dom. in Monte), in the words, "Your reward is very great in heaven" (Mat. 5:12).

Again, three kinds of supernatural visions, bodily, imaginative, and intellectual, are called sometimes so many heavens, in reference to which Augustine (Gen. ad lit. xii) expounds Paul's rapture "to the third heaven."

Reply to Objection 1: The earth stands in relation to the heaven as the centre of a circle to its circumference. But as one center may have many circumferences, so, though there is but one earth, there may be many heavens.

Reply to Objection 2: The argument holds good as to the heaven, in so far as it denotes the entire sum of corporeal creation, for in that sense it is one.

Reply to Objection 3: All the heavens have in common sublimity and some degree of luminosity, as appears from what has been said.

ON THE WORK OF THE THIRD DAY (TWO ARTICLES)

We next consider the work of the third day. Under this head there are two points of inquiry:

(1) About the gathering together of the waters;

(2) About the production of plants.

Whether it was fitting that the gathering together of the waters should take place, as recorded, on the third day?

Objection 1: It would seem that it was not fitting that the gathering together of the waters should take place on the third day. For what was made on the first and second days is expressly said to have been "made" in the words, "God said: Be light made," and "Let there be a firmament made."But the third day is contradistinguished from the first and the second days. Therefore the work of the third day should have been described as a making not as a gathering together.

Objection 2: Further, the earth hitherto had been completely covered by the waters, wherefore it was described as "invisible" [*Q[66], A[1], OBJ[1]]. There was then no place on the earth to which the waters could be gathered together.

Objection 3: Further, things which are not in continuous contact cannot occupy one place. But not all the waters are in continuous contact, and therefore all were not gathered together into one place.

Objection 4: Further, a gathering together is a mode of local movement. But the waters flow naturally, and take their course towards the sea. In their case, therefore, a Divine precept of this kind was unnecessary.

Objection 5: Further, the earth is given its name at its first creation by the words, "In the beginning God created heaven and earth." Therefore the imposition of its name on the third day seems to be recorded without necessity.

On the contrary, The authority of Scripture suffices.

I answer that, It is necessary to reply differently to this question according to the different interpretations given by Augustine and other holy writers. In all these works, according to Augustine (Gen. ad lit. i, 15; iv, 22,34; De Gen. Contr. Manich. i, 5, 7), there is no order of duration, but only of origin and nature. He says that the formless spiritual and formless corporeal natures were created first of all, and that the latter are at first indicated by the words "earth" and "water." Not that this formlessness preceded formation, in time, but only in origin; nor yet that one formation preceded another in duration, but merely in the order of nature. Agreeably, then, to this order, the formation of the highest or spiritual nature is recorded in the first place, where it is said that light was made on the first day. For as the spiritual nature is higher than the corporeal, so the higher bodies are nobler than the lower. Hence the formation of the higher bodies is indicated in the second place, by the words, "Let there be made a firmament," by which is to be understood the impression of celestial forms on formless matter, that preceded with priority not of time, but of origin only. But in the third place the impression of elemental forms on formless matter is recorded, also with a priority of origin only. Therefore the words, "Let the waters be gathered together, and the dry land appear," mean that corporeal matter was impressed with the substantial form of water, so as to have such movement, and with the substantial form of earth, so as to have such an appearance.

According, however, to other holy writers [*Q[66], A[1]] an order of duration in the works is to be understood, by which is meant that the formlessness of matter precedes its formation, and one form another, in order of time. Nevertheless, they do not hold that the formlessness of matter implies the total absence of form, since heaven, earth, and water already existed, since these three are named as already clearly perceptible to the senses; rather they understand by formlessness the want of due distinction and of perfect beauty, and in respect of these three Scripture mentions three kinds of formlessness. Heaven, the highest of them, was without form so long as "darkness" filled it, because it was the source of light. The formlessness of water, which holds the middle place, is called the "deep," because, as Augustine says (Contr. Faust. xxii, 11), this word signifies the mass of waters without order. Thirdly, the formless state of the earth is touched upon when the earth is

said to be "void" or "invisible," because it was covered by the waters. Thus, then, the formation of the highest body took place on the first day. And since time results from the movement of the heaven, and is the numerical measure of the movement of the highest body, from this formation, resulted the distinction of time, namely, that of night and day. On the second day the intermediate body, water, was formed, receiving from the firmament a sort of distinction and order (so that water be understood as including certain other things, as explained above (Q[68], A[3])). On the third day the earth, the lowest body, received its form by the withdrawal of the waters, and there resulted the distinction in the lowest body, namely, of land and sea. Hence Scripture, having clearly expresses the manner in which it received its form by the equally suitable words, "Let the dry land appear."

Reply to Objection 1: According to Augustine [*Gen. ad lit. ii, 7,8; iii, 20], Scripture does not say of the work of the third day, that it was made, as it says of those that precede, in order to show that higher and spiritual forms, such as the angels and the heavenly bodies, are perfect and stable in being, whereas inferior forms are imperfect and mutable. Hence the impression of such forms is signified by the gathering of the waters, and the appearing of the land. For "water," to use Augustine's words, "glides and flows away, the earth abides" (Gen. ad lit. ii, 11). Others, again, hold that the work of the third day was perfected on that day only as regards movement from place to place, and that for this reason Scripture had no reason to speak of it as made.

Reply to Objection 2: This argument is easily solved, according to Augustine's opinion (De Gen. Contr. Manich. i), because we need not suppose that the earth was first covered by the waters, and that these were afterwards gathered together, but that they were produced in this very gathering together. But according to the other writers there are three solutions, which Augustine gives (Gen. ad lit. i, 12). The first supposes that the waters are heaped up to a greater height at the place where they were gathered together, for it has been proved in regard to the Red Sea, that the sea is higher than the land, as Basil remarks (Hom. iv in Hexaem.). The second explains the water that covered the earth as being rarefied or nebulous, which was afterwards condensed when the waters were gathered together. The third suggests the existence of hollows in the earth, to receive the confluence of waters. Of the above the first seems the most probable.

Reply to Objection 3: All the waters have the sea as their goal, into which they flow by channels hidden or apparent, and this may be the reason why they are said to be gathered together into one place. Or, "one place" is to be understood not simply, but as contrasted with the place of the dry land, so that the sense would be, "Let the waters be gathered together in one place," that is, apart from the dry land. That the waters occupied more places than one seems to be implied by the words that follow, "The gathering together of the waters He called Seas."

Reply to Objection 4: The Divine command gives bodies their natural movement and by these natural movements they are said to "fulfill His word." Or we may say that it was according to the nature of water completely to cover the earth, just as the air completely surrounds both water and earth; but as a necessary means towards an end, namely, that plants and animals might be on the earth, it was necessary for the waters to be withdrawn from a portion of the earth. Some philosophers attribute this uncovering of the earth's surface to the action of the sun lifting up the vapors and thus drying the land. Scripture, however, attributes it to the Divine power, not only in the Book of Genesis, but also Job 38:10 where in the person of the Lord it is said, "I set My bounds around the sea," and Jer. 5:22, where it is written: "Will you not then fear Me, saith the Lord, who have set the sand a bound for the sea?"

Reply to Objection 5: According to Augustine (De Gen. Contr. Manich. i), primary matter is meant by the word earth, where first mentioned, but in the present passage it is to be taken for the element itself. Again it may be said with Basil (Hom. iv in Hexaem.), that the earth is mentioned in the first passage in respect of its nature, but here in respect of its principal property, namely, dryness. Wherefore it is written: "He called the dry land, Earth." It may also be said with Rabbi Moses, that the expression, "He called," denotes throughout an equivocal use of the name imposed. Thus we find it said at first that "He called the light Day": for the reason that later on a period of twenty-four hours is also called day, where it is said that "there was evening and morning, one day." In like manner it is said that "the firmament," that is, the air, "He called heaven": for that which was first created was also called "heaven." And here, again, it is said that "the dry land," that is, the part from which the waters had withdrawn, "He called, Earth," as distinct from the sea; although the name earth is equally applied to that which is covered with waters or not. So by the expression "He called" we are to understand throughout that the nature or property He bestowed corresponded to the name He gave.

Whether it was fitting that the production of plants should take place on the third day?

Objection 1: It would seem that it was not fitting that the production of plants should take place on the third day. For plants have life, as animals have. But the production of animals belongs to the work, not of distinction, but of adornment. Therefore the production of plants, as also belonging to the work of adornment, ought not to be recorded as taking place on the third day, which is devoted to the work of distinction.

Objection 2: Further, a work by which the earth is accursed should have been recorded apart from the work by which it receives its form. But the words of Gn. 3:17, "Cursed is the earth in thy work, thorns and thistles shall it bring forth to thee," show that by the production of certain plants the earth was accursed. Therefore the production of plants in general should not have been recorded on the third day, which is concerned with the work of formation.

Objection 3: Further, as plants are firmly fixed to the earth, so are stones and metals, which are, nevertheless, not mentioned in the work of formation. Plants, therefore, ought not to have been made on the third day.

On the contrary, It is said (Gn. 1:12): "The earth brought forth the green herb," after which there follows, "The evening and the morning were the third day."

I answer that, On the third day, as said (A[1]), the formless state of the earth comes to an end. But this state is described as twofold. On the one hand, the earth was "invisible" or "void," being covered by the waters; on the other hand, it was "shapeless" or "empty," that is, without that comeliness which it owes to the plants that clothe it, as it were, with a garment. Thus, therefore, in either respect this formless state ends on the third day: first, when "the waters were gathered together into one place and the dry land appeared"; secondly, when "the earth brought forth the green herb." But concerning the production of plants, Augustine's opinion differs from that of others. For other commentators, in accordance with the surface meaning of the text, consider that the plants were produced in act in their various species on this third day; whereas Augustine (Gen. ad lit. v, 5; viii, 3) says that the earth is said to have then produced plants and trees in their causes, that is, it received then the power to produce them. He supports this view by the authority of Scripture, for it is said

(Gn. 2:4,5): "These are the generations of the heaven and the earth, when they were created, in the day that . . . God made the heaven and the earth, and every plant of the field before it sprung up in the earth, and every herb of the ground before it grew." Therefore, the production of plants in their causes, within the earth, took place before they sprang up from the earth's surface. And this is confirmed by reason, as follows. In these first days God created all things in their origin or causes, and from this work He subsequently rested. Yet afterwards, by governing His creatures, in the work of propagation, "He worketh until now."Now the production of plants from out the earth is a work of propagation, and therefore they were not produced in act on the third day, but in their causes only. However, in accordance with other writers, it may be said that the first constitution of species belongs to the work of the six days, but the reproduction among them of like from like, to the government of the universe. And Scripture indicates this in the words, "before it sprung up in the earth," and "before it grew," that is, before like was produced from like; just as now happens in the natural course by the production of seed. Wherefore Scripture says pointedly (Gn. 1:11): "Let the earth bring forth the green herb, and such as may seed," as indicating the production of perfection of perfect species, from which the seed of others should arise. Nor does the question where the seminal power may reside, whether in root, stem, or fruit, affect the argument.

Reply to Objection 1: Life in plants is hidden, since they lack sense and local movement, by which the animate and the inanimate are chiefly discernible. And therefore, since they are firmly fixed in the earth, their production is treated as a part of the earth's formation.

Reply to Objection 2: Even before the earth was accursed, thorns and thistles had been produced, either virtually or actually. But they were not produced in punishment of man; as though the earth, which he tilled to gain his food, produced unfruitful and noxious plants. Hence it is said: "Shall it bring forth TO THEE."

Reply to Objection 3: Moses put before the people such things only as were manifest to their senses, as we have said (Q[67], A[4]; Q[68], A[3]). But minerals are generated in hidden ways within the bowels of the earth. Moreover they seem hardly specifically distinct from earth, and would seem to be species thereof. For this reason, therefore, he makes no mention of them.

OF THE WORK OF ADORNMENT, AS REGARDS THE FOURTH DAY (THREE ARTICLES)

We must next consider the work of adornment, first as to each day by itself, secondly as to all seven days in general.

In the first place, then, we consider the work of the fourth day, secondly, that of the fifth day, thirdly, that of the sixth day, and fourthly, such matters as belong to the seventh day.

Under the first head there are three points of inquiry:

(1) As to the production of the lights;

(2) As to the end of their production;

(3) Whether they are living beings?

Whether the lights ought to have been produced on the fourth day?